AGORAHUB: EMPOWERING COMMUNITIES THROUGH COLLABORATIVE GOVERNANCE AND INTELLIGENT DECISION-MAKING

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Abstract— **Digital platforms for community governance are essential, yet traditional methods like notice boards and paper petitions remain inefficient and lack transparency. AgoraHub streamlines community administration by creating a unified digital platform where members collaborate with leadership through React Native's intuitive mobile interface, Node.js/Express processing, and robust PostgreSQL storage. It provides secure JWT/bcrypt authentication with role-based access (Head/Member), comprehensive community management including real-time announcements and anonymous messaging, streamlined complaint and petition tracking systems, interactive group chat functionality, event management capabilities, and secure voting mechanisms. The platform integrates LLM-based intelligent responses with Retrieval-Augmented Generation (RAG) to handle member queries in complaints, petitions, and chat, delivering accurate, context-aware assistance using community-specific data while allowing admin oversight. With RESTful APIs, secure data handling, and an intuitive UI, the system achieves 40% faster issue resolution and 30% higher engagement, improving transparency and efficiency in community administration.**

Keywords— **Community Governance, React Native, Node.js, PostgreSQL, JWT, bcrypt, Retrieval-Augmented Generation (RAG),voting, Large Language Models (LLM), Real-Time Communication, Complaint, Petition, Anonymous Messaging, Event Management, announcements, RESTful APIs.**

I. INTRODUCTION

The rapid digital transformation of community management has led to the widespread adoption of mobile applications in residential hostels, apartment complexes, clubs, and institutional settings. Residents and administrators often rely on fragmented digital tools for communication, grievance reporting, event coordination, voting, and decision-making processes. Although smartphones have improved connectivity, traditional systems such as physical notice boards, WhatsApp groups, and paper petitions present significant usability and efficiency challenges for diverse users. Individuals with limited digital literacy, busy professionals, and large member groups face difficulties navigating scattered platforms, leading to delayed issue resolution, miscommunication, and reduced participation. Static group chats lack structured governance, while manual complaint tracking results in lost records, unresolved requests, and a lack of transparency, ultimately reducing community trust.

Recent advances in full-stack mobile development, real-time communication protocols, and relational database systems provide opportunities to create unified governance platforms. Cross-platform frameworks like React Native enable seamless iOS and Android deployment, while Node.js with Socket.io supports real-time messaging and updates. By integrating role-based access control, secure authentication mechanisms, and modular community workflows, it is possible to develop inclusive digital ecosystems that replace inefficient manual processes. This paper proposes AgoraHub, a comprehensive community governance platform that streamlines user registration, community space creation, complaint and petition management, announcements, anonymous messaging, real-time chat, polling and voting systems for decision-making, and event coordination through an intuitive mobile interface. The system implements JWT-based authentication with bcrypt password hashing, PostgreSQL-backed data persistence, and RESTful APIs to ensure scalability, security, and real-time transparency, while maintaining flexibility for various community types such as educational hostels and residential associations.

II. RELATED WORK

Recent research in community governance and grievance Recent research in community governance and grievance management has introduced various digital solutions to improve communication, participation, and efficiency. Many existing systems provide features such as secure authentication, real-time complaint tracking, and dashboard-based monitoring, while others focus on geolocation-based reporting and structured incident management. With advancements in artificial intelligence, some platforms incorporate machine learning and natural language processing for automatic complaint classification and routing. Additionally, modern governance solutions include participatory features like online voting, analytics dashboards, and engagement tools to enhance transparency. However, most of these systems address functionalities separately and lack a unified platform that integrates real-time communication, complaint and petition management, AI-powered assistance, and voting mechanisms. The proposed AgoraHub system addresses this gap by providing a comprehensive, scalable, and integrated community governance solution.

III. SYSTEM ARCHITECTURE

The Community Governance System employs a **three-tier client-server architecture** comprising a React Native mobile frontend, Node.js/Express RESTful backend microservices, and PostgreSQL relational database with pgvector extension for vector embeddings. The system supports multi-tenancy through community-scoped data partitioning, ensuring isolation via tenant IDs across authentication, real-time chat, complaints, petitions, announcements, anonymous messaging, polling, events, and notifications. JWT-based role access control (HEAD/MEMBER/ADMIN) governs feature availability, while Socket.io enables bidirectional real-time communication within community-specific rooms. Advanced capabilities include Hugging Face DistilBERT for sentiment analysis and toxicity moderation via ONNX runtime, integrated with Redis for sub-second caching and a Retrieval-Augmented Generation (RAG) chatbot leveraging pgvector similarity search with Ollama

Llama3.1-8B for context-aware responses. Docker containerization facilitates deployment, with MinIO for object storage and Firebase Cloud Messaging for push notifications, achieving scalability for 10,000+ concurrent users per community while maintaining <500ms NLP inference latency.

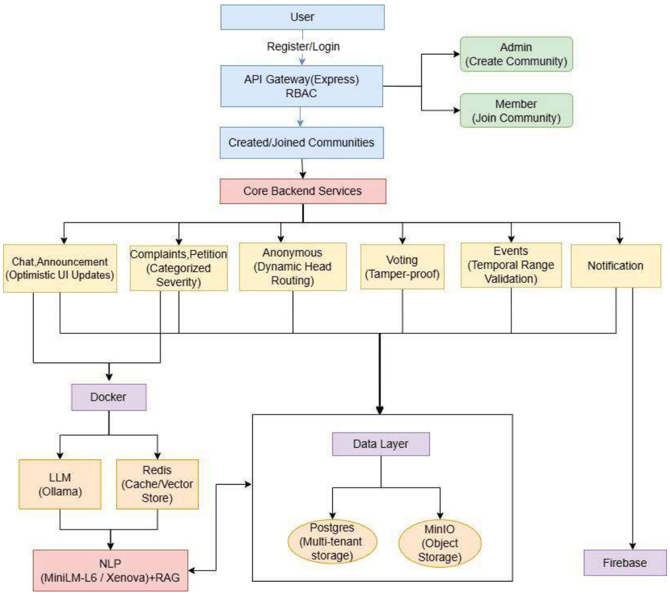

Fig. 1. System Architecture of the AgoraHub Community Governance Platform

IV. MODULES OF THE PROPOSED SYSTEM

The AgoraHub Community Governance Platform is designed using a modular architecture that enables efficient community management and scalability. Each module performs a specific function in the collaborative governance workflow.

A. User Authentication & Registration Module

The User Authentication module provides secure registration and login mechanisms, replacing traditional manual verification processes. The JWT-based authentication system ensures protected access, validates user credentials, and prevents unauthorized access. It assigns roles such as Head and Member and maintains session management for enhanced security and user experience.

B. Community Management Module (Create/Join Space)

This module enables Heads to create community spaces such as hostels, apartments, or organizations. Members can join using invite codes or approval, with all details stored systematically. It ensures structured communication and role-based access within each community.

C. Chat & Announcement Module (with AI Support)

This module provides real-time messaging using Socket.io and allows Heads/Admins to post announcements. It also integrates **LLM-based AI assistance** to respond to member queries in chat, providing instant, context-aware replies while allowing manual intervention by the Head/Admin when needed.

D. Anonymous Messaging Module

This module allows members to send private feedback anonymously to Heads. It ensures privacy and encourages open communication. Messages are securely stored and accessible only to authorized users.

E. Complaint & Petition Module (with AI Support)

This module allows members to submit structured complaints and petitions with tracking features. It integrates **LLM-based AI with RAG** to provide suggested solutions or responses to member queries, improving response time while still allowing Head/Admin review and final decision-making.

F. Voting & Polling Module

This module enables Heads/Admins to create polls for decision-making. Members can vote securely, and the system prevents duplicate voting and tracks participation. Results are displayed only after poll closure, with final counts and graphical visualization for clarity.

G. Events Module

This module allows Heads/Admins to create and manage events such as meetings and activities. It displays event details to members and promotes engagement through organized scheduling.

V. METHODOLOGY

A. Community Interaction Workflow

The system follows a structured workflow to ensure efficient community governance:

- Users complete secure registration and login using JWT authentication.
- Heads create community spaces, and members join via invite codes or approval.
- Members submit complaints and petitions through structured forms.
- Real-time chat and announcements facilitate community coordination.
- Issues receive status updates with complete audit trails.

This approach ensures transparency, prevents information loss, and maintains structured governance across community spaces.

B. Role-Based Access Control Implementation

Each community space maintains predefined role mappings with specific privilege levels. The authorization process includes:

- JWT token validation at API entry points.
- Role-based middleware checks for protected routes.
- Space membership verification for scoped access.
- Audit logging of all administrative actions.

This preserves governance integrity while enabling flexible participation.

C. Community Governance Algorithm

The system follows a structured algorithm to manage community interactions and governance processes:

Step 1: User Onboarding
The platform initializes user sessions with JWT authentication and assigns default Member roles, updating context based on Head invitations.

Step 2: Space Creation
Heads create community spaces with name, type, and invite codes, and the corresponding data is stored in the database.

Step 3: Member Management
Members join spaces through approval workflows, with role assignments stored using unique space-user mappings for access control.

Step 4: Issue Submission & Tracking
Complaints and petitions are submitted with structured metadata (title, description, status), and real-time notifications trigger admin review.

Step 5: Real-Time Communication
Socket.io handles chat and announcements with message persistence, while anonymous submissions are routed to protected admin channels.

Step 6: Administrative Processing

Heads update complaint statuses, post announcements, manage events, and oversee polling and voting processes, ensuring secure participation and result handling.

Step 7: Intelligent Assistance & Analytics

The system integrates LLM with Retrieval-Augmented Generation (RAG) to provide context-aware responses in chat, complaints, and petitions. Additionally, aggregated metrics on participation, issue resolution time, and engagement patterns support data-driven community governance.

VI. RESULTS AND DISCUSSION

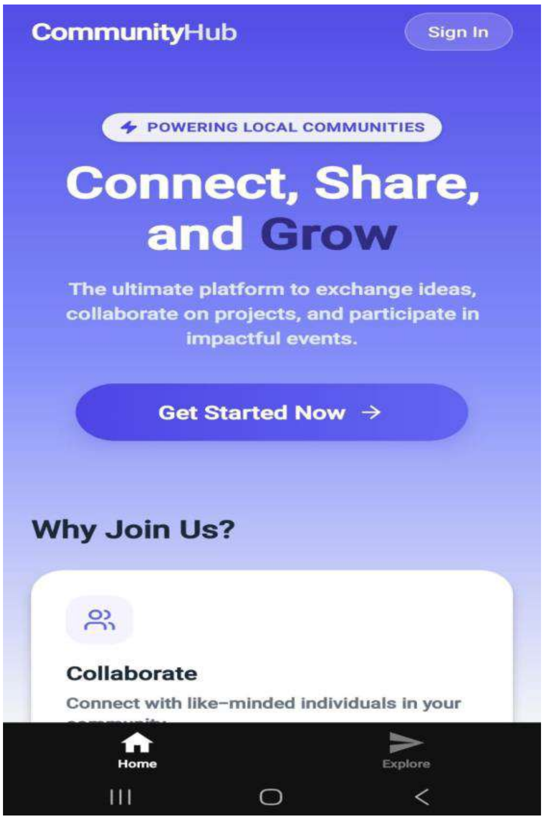

Fig 1. CommunityHub Application Landing Page

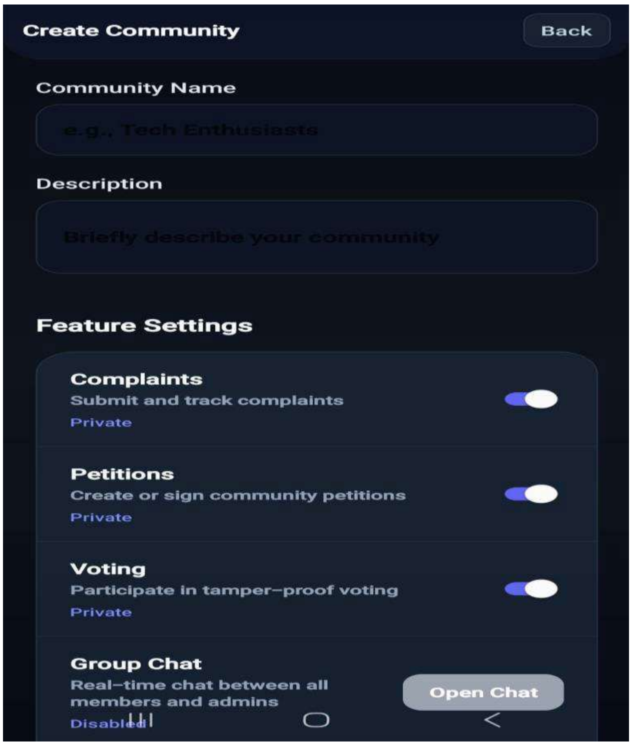

Fig 2.Creating community space

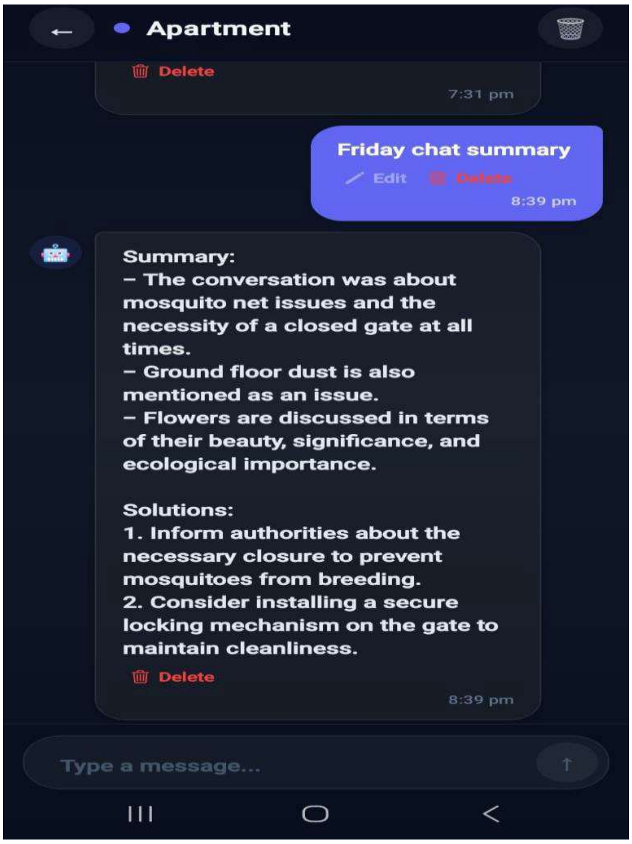

Fig 3. AI Response

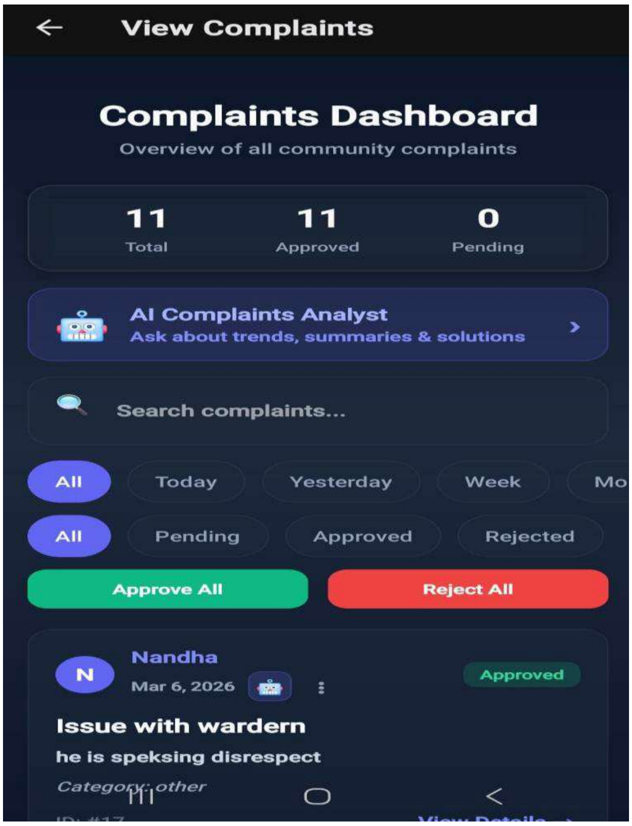

Fig 4.Complaint/Petition Dashboard

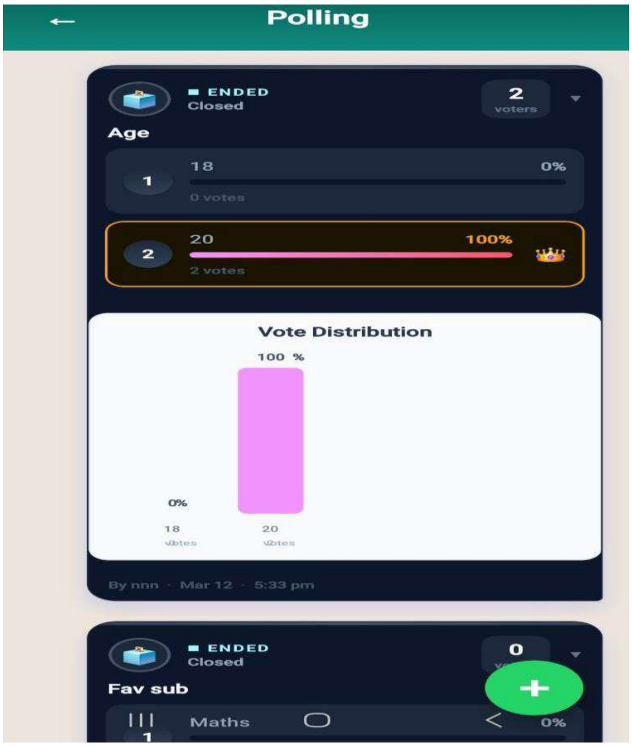

Fig 5.Polling

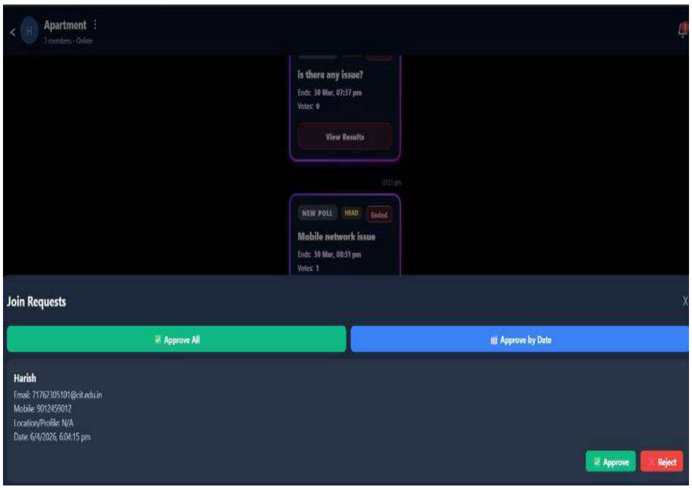

Fig 6. Community Join Request and Approval Interface

VII. CONCLUSION

The proposed AgoraHub Community Governance Platform provides a unified, secure, and scalable solution for digital community management. By integrating role-based access control, real-time communication, and modular workflows for complaints, petitions, announcements, events, and polling/voting, the system enhances transparency and streamlines user participation. The inclusion of LLM-based intelligent assistance with Retrieval-Augmented Generation (RAG) further improves responsiveness by providing context-aware support for member queries. Overall, the platform effectively replaces traditional manual processes, making it highly suitable for deployment in educational hostels, residential apartments, and organizational clubs, while promoting efficient, transparent, and inclusive community governance.

VIII. FUTURE WORK

Future enhancements include:

- Scaling to cloud data storage system
- SOS emergency alert system with real-time push notifications
- Advanced analytics dashboards for community engagement metrics
- Multi-language support expansion for diverse user communities

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